

TIC 318620649

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 13.834347 d (ALIASED, see below)
-> corrected period: 27.668693 d
Epoch (t0) 1960.0539 BTJD
Duration 0.2440 d
Depth 0.919 % (9,190 ppm)
SNR 60.8

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

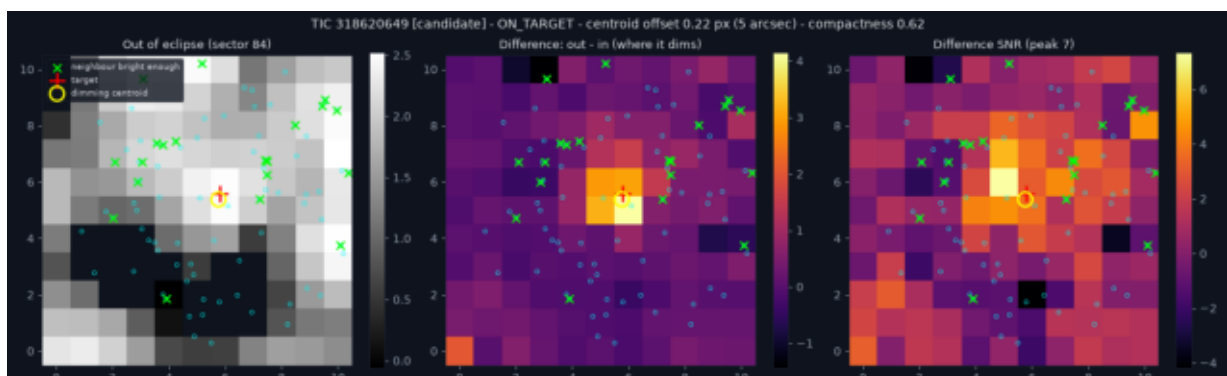
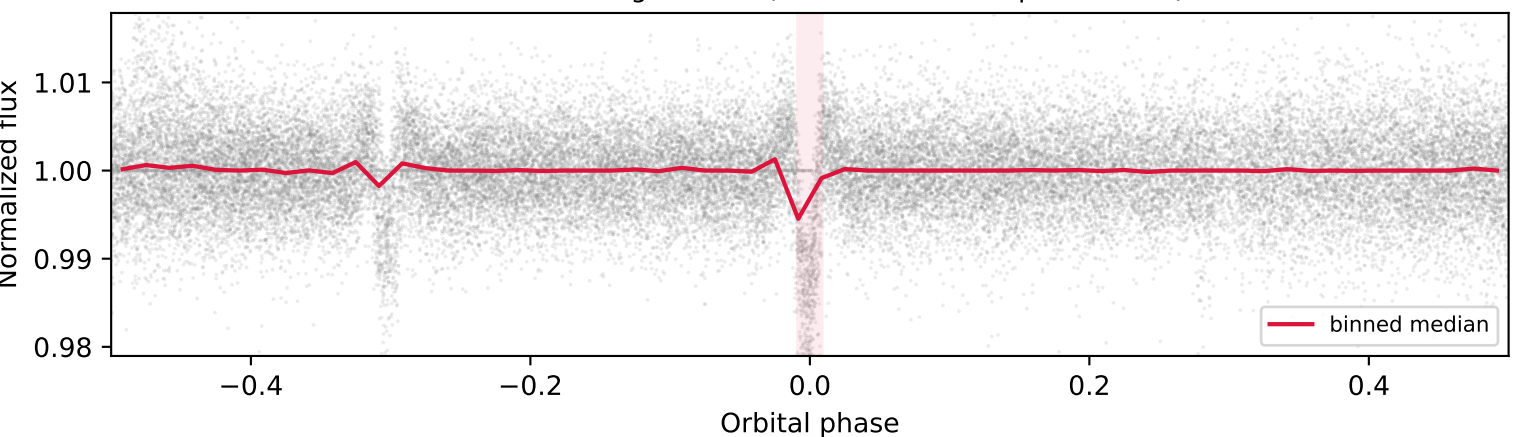
Eclipses observed 9 (gate ≥ 3)
Robust depth ratio 1.00 (gate ≥ 0.50)
Robust depth sigma 46.2 (gate ≥ 7.0)
Odd/even z-score 4.33 (ALIAS FLAGGED)
Secondary eclipse detected, phase -0.30, 7.0 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 84 light-curve dip in this sector: 19.1 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 7.34 (gate ≥ 5.0) Compactness 0.62 (gate ≥ 0.50)
Dimming centroid offset 0.22 px (5 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 318620662
1.45 px away, dTmag +3.82 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 4.33 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 27.668693 d, in any write-up.